Finite Fields, Chevally's Theorem and Erdos-Ginzburg-Ziv's Theorem

Recall the *congruence* (*residue*) *classes of modulus m*, $\mathbb{Z}/m\mathbb{Z} = \{0,1,2,...,m-1\}$. With usual addition and multiplication modulo *m*, the set forms a *ring*. In particular if *m* is a prime *p*, then we have $\mathbb{Z}/p\mathbb{Z}$, then the set forms a *field*. Mainly every nonzero element has an inverse. We know some properties concerning this field, say Fermat's little theorem, Wilson's theorem, or Wolstenholme's. Here we aim to produce something new (*finite fields*) from these $\mathbb{Z}/p\mathbb{Z}$.

Example (F_{2^n}). We start from $F_2 = \mathbb{Z}/2\mathbb{Z}$. It is a field with two elements. Now we consider polynomials *irreducible* in this field, namely we look for polynomials of *degree* ≥ 2 which cannot be factored into smaller components. There are three polynomials of degree 2, $x^2, x^2+1=(x+1)^2$ and x^2+x+1. The last one is irreducible, for neither 0 nor 1 is a root. Now *imagine* there is an α which satisfies $\alpha^2+\alpha+1=0$. We *add* (*adjoin*) this element to the original field, we get 4 elements, $a+b\alpha$, with $a,b=0,1$. These four elements actually form a field containing the original one, and the field is actually F_{2^2}. It is also a *vector space of dimension* 2 over F_2. One more thing, we have $\alpha, \alpha^2=\alpha+1$ and finally $\alpha^3=\alpha\cdot\alpha^2=\alpha(\alpha+1)=\alpha^2+\alpha=1$. Hence powers of α generate all nonzero elements. Likewise $1+\alpha, (1+\alpha)^2=1+\alpha^2=1+\alpha+1=\alpha$ and finally $(1+\alpha)^3=1$. Thus α and $1+\alpha$ are *primitive* elements.

The irreducible polynomial can be obtained by factoring

$$x^{2^2}-x=x^4-x=x(x^3-1)=x(x-1)(x^2+x+1).$$

Actually this *splitting* polynomial contains all the elements in F_{2^2}.

By the same token, consider

$$x^{2^3}-x=x^8-x=x(x^7-1)=x(x-1)(x^6+x^5+x^4+x^3+x^2+x+1)$$
$$=x(x-1)(x^3+x+1)(x^3+x^2+1).$$

Take β as a root of $x^3+x+1=0$, then we have altogether 8 elements generated $a+b\beta+c\beta^2$, with $a,b,c=0,1$. Taking powers of β, we have respectively $\beta,\beta^2,\beta^3=\beta+1,\beta^4=\beta^2+\beta,\beta^5=\beta^3+\beta^2=\beta^2+\beta+1,\beta^6=\beta^2+1$ and finally $\beta^7=1$. It is a primitive element. Using the powers of β, we find the roots of

$x^3+x+1=0$ are β, β^2 and β^4. While the roots of $x^3+x^2+1=0$ are β^3, β^5 and β^6. Both cubic polynomials are *primitive polynomials*, their roots generate all nonzero elements.

Example: (F_{3^n}). Now attention now drawn to extension of $F_3 = \mathbb{Z}/3\mathbb{Z}$. Because computations are getting complicated, we just consider

$$\begin{aligned}
x^{3^2} - x &= x^9 - x = x(x^8-1) = x(x^4-1)(x^4+1) \\
&= x(x^2-1)(x^2+1)(x^4+1) \\
&= x(x^2-1)(x^2+1)(x^4+1) \\
&= x(x-1)(x+1)(x^2+1)(x^4+1) \\
&= x(x-1)(x+1)(x^2+1)(x^2+x+2)(x^2+2x+2).
\end{aligned}$$

Choose i as a root of $x^2+1=0$, then by adjoining i, we have 9 elements, $a+bi$, with $a,b=0,1,2$. Note i is of order 4, hence not a primitive. If we consider $\alpha = 1+i$, then it is a primitive, by taking powers of α, we have respectively $\alpha^2 = 2i = -i, \alpha^3 = 1-i, \alpha^4 = -1, \alpha^5 = -1-i, \alpha^6 = i, \alpha^7 = -1+i$ and finally $\alpha^8 = 1$. If we take γ a root of $x^2+x+2=0$, then we can check γ generates all nonzero elements. Using γ, we check γ, γ^3 are roots of $x^2+x+2=0$, γ^5, γ^7 are roots of $x^2+x+2=0$. (Naturally as they are primitive polynomials.) On the other hand, γ^2, γ^6 are roots of $x^2+1=0$ (not a primitive polynomial). Finally γ^4 is the root of $x+1=0$ and $\gamma^8=1$ is the root of $x-1=0$.

Example: Finite fields may be written as a ring modulo an irreducible polynomial $F[x]/(f(x))$. For examples, $F_2[x]/(x^3+x+1)$ and $F_2[x]/(x^3+x^2+1)$ are two (isomorphic) fields of order 8, $F_3[x]/(x^2+1)$ and $F_3[x]/(x^2+x+1)$ are two isomorphic fields of order 9. In the first case, x is of order 4 (not a primitive); in the second case, x is of order 8 (a primitive). Another example is $F_7[x]/(x^3-2)$ is a field of order $7^3 = 243$.

Finite field may appear in different forms.

Example (F_{3^n}). Consider $\mathbb{Z}\left[\sqrt{2}\right]/(3)$. Its elements are $a+b\sqrt{2}$, with $a,b=0,1,2$. Elements of order 1 is 1, order 2 is 2, order 4 are $\sqrt{2}$ and $2\sqrt{2}$, order 8 are $\alpha = 1+\sqrt{2}$, $\alpha^3 = 1+2\sqrt{2}$, $\alpha^5 = 2+2\sqrt{2}$ and $\alpha^7 = 2+\sqrt{2}$ (primitives).

We now just state some basic properties of finite fields. Prime field, characteristic etc Starting from the *ground* (*prime*) field $F_p = \mathbb{Z}/p\mathbb{Z}$, we use an *irreducible* polynomial $f(x)$ of degree n to build a field of order (size) p^n, denoted by F_{p^n} or $\mathbb{Z}[x]/(f(x))$. The prime p is called the *characteristic* of the finite field, it is also the smallest number of 1 that add up to 0, clearly it is a prime number. We just list several properties.

1. For a prime p and a monic irreducible polynomial $f(x)$ in $F_p[x]$ of degree n, the ring $F_p[x]/(f(x))$ is a field of order p^n.
2. For every prime power p^n, a finite field of order p^n exists.
3. A field of prime power of order p^n is a splitting field over F_p of $x^{p^n} - x$.
4. Any finite fields of the same size are isomorphic.
5. A subfield of F_{p^d} has order p^d where $d|n$, and there is one such field for each d.

We save ourselves from talking about Galois theory.

One more basic fact:

<u>Theorem</u>: If F $(|F| = p^n = q)$ is a finite field, then the group of nonzero (invertible) elements F^* is cyclic.

First if $a_1, a_2, ..., a_{q-1}$, where $q = p^n$ are all the nonzero elements of F, multiplying by a nonzero element b, we get a permutation of them. Thus by taking products and cancelling out the a_i's, we get $b^{q-1} = 1$. As b is arbitrary, we have $a^{q-1} = 1$ for all nonzero elements, or $a^q = a$ for all elements. Now if a is a nonzero element and d (*order*) is the smallest element such that $a^d = 1$, then $d \mid (q-1)$. Indeed $q-1 = \lambda d + r, 0 \le r < d$, then $1 = a^{q-1} = a^{\lambda d + r} = a^{dr} a^r = a^r$, by the minimality of *d*, we have $r = 0$. We proceed to show there exists *a* such that its order is $q-1$, by giving several proofs.

(Elementary) Assume $q \ge 3$, set $h = q - 1 = \left|F^*\right|$ be the size of nonzero elements, now let $h = p_1^{r_1} p_2^{r_2} \dots p_m^{r_m}$ be its prime factorization. Now for each $i, 1 \le i \le m$, the polynomial $x^{h/p_i} - 1$ has at most h / p_i roots in *F*, since $h / p_i < h$, there is a nonzero element a_i which is not a root of the equation $x^{h/p_i} - 1 = 0$. Set $b_i = a_i^{h/p_i^{r_i}}$. Then $b_i^{p_i^{r_i}} = 1$, so the order of b_i is of the form $p_i^{s_i}$, where $0 \le s_i \le r_i$. But on the other hand, $b_i^{p_i^{r_i - 1}} = a_i^{h/p_i} \ne 1$, so the order of b_i is exactly $p_i^{r_i}$. Finally take $b = b_1 b_2 \dots b_m$, then its order is exactly $h = q - 1$, (a generator or primitive). Suppose not, then its order is a divisor of at least one of the integers h / p_i, say h / p_1. Then $1 = b^{h/p_1} = b_1^{h/p_1} b_2^{h/p_1} \dots b_m^{h/p_1}$. Now for $2 \le i \le m$, $p_i^{r_i}$ divides h / p_1, and so $b_i^{h/p_1} = 1$. This forces $b_1^{h/p_1} = 1$, which is impossible since the order of b_1 is exactly $p_1^{r_1}$.

(Number theoretic) Let *a* be an element of order *d* $(d \mid q-1)$ then $a, a^2, \dots, a^d = 1$ are distinct elements satisfying $x^d - 1 = 0$ in F. By the theorem of Lagrange, these are all elements of orders that divide d in *F*. Now if *j* is not relatively prime to *d*, then a^j is of order $< d$. For we can let $1 < d' = \gcd(j, d)$, then $(a^j)^{d/d'} = 1$, hence its order is strictly smaller than *d*. If $\gcd(j, d) = 1$, we know there exists λ, μ such that $\lambda j + \mu d = 1$. Suppose a^j is of order $d'' < d$, then
$a^{d''} = a^{d''(\lambda j + \mu d)} = (a^j)^{d'' \lambda} (a^d)^{u d''} = 1$, hence a is of order d'' strictly less than *d*, a contradiction. Thus among the elements, there are exactly $\phi(d)$ (Euler phi function) of them whose order are exactly *d*. Thus for each divisor *d*, either there are no element of order *d*, or there are $\phi(d)$ of them. However we know also
$\sum_{d \mid q-1} \phi(d) = q - 1$, which are the number of elements in F^*. To complete the picture, there exist exactly $\phi(q-1)$ elements of order $q-1$.

(Group theoretic) Let $q = p^n = |F|$, then the size of nonzero elements

$|F^*| = p^n - 1 = q - 1$. Let m be the maximal order of the elements of F^*, then

$m|(q-1)$, by Lagrange's theorem. On the other hand, from finite abelian (commutative) group theory, the order of every element divides the maximal order, thus every element satisfies $x^m = 1$. Therefore all elements are roots of $x^m - 1 = 0$. Again from a theorem of Lagrange, the equation has at most m roots, thus we must have $q - 1 \le m$. Combining with $m|(q-1)$, we must have $m = q - 1$.

<u>Remark</u>: Elements of maximal order are called *primitives*, it is not necessarily easy to find them, though we know they are there.

<u>Warning</u>: Distinguish between finite fields F_{p^n} with $p^n - 1$ nonzero (invertible) elements from invertible elements in the residue classes, $(\mathbb{Z}/p^n\mathbb{Z})^*$, there are $p^n - p^{n-1}$ of them. Theorem of Gauss: $(\mathbb{Z}/m\mathbb{Z})^*$ containing primitives if and only if $m = 2, 4, p^\alpha, 2p^\alpha$, with p an odd prime. For instance $(\mathbb{Z}/8\mathbb{Z})^*$ has four invertible elements $1, 3, 5, 7$, all are of order 2 and no primitive.

We now describe a theorem of Chevally. First the *degree* of a monomial of k variables $x_1^{\alpha_1} x_2^{\alpha_2} ... x_k^{\alpha_k}$ is the sum of nonnegative integers $\alpha_1 + \alpha_2 + ... + \alpha_k$. The degree of a sum of monomials (polynomial) is the largest possible degree of the monomials. All the polynomials of k variables with coefficients from the field F form a ring, denoted by $F[x_1, x_2, ..., x_k]$.

Now the theorem of Chevally-Warning:

<u>Theorem</u>: Let $F = F_q = F_{p^n}$ be a finite field and $f \in F[x_1, x_2, ..., x_n]$, the degree of f is strictly less than n. Let N be the number of solutions of $f(x_1, x_2, ..., x_n) = 0$ in F_q.

Then $p|N$.

We need two lemmas.

Lemma 1: Let $k \geq 0$, then

$$\sum_{c \in F} c^k = \begin{cases} 0, & \text{if } k = 0 \text{ or } q-1 \nmid k; \\ -1 & \text{if } k > 0 \text{ and } q-1 \mid k. \end{cases}$$

Proof: We define $0^0 = 1$ and the case for $k = 0$ is clearly true. Now assume $k > 0$, pick a *primitive* of *F*, then

$$\sum_{c \in F} c^k = \sum_{c \in F^*} c^k = \sum_{j=0}^{q-2} (g^j)^k = \sum_{j=0}^{q-2} (g^k)^j$$

$$= \begin{cases} \dfrac{(g^k)^{q-1} - 1}{g^k - 1} = 0, & \text{if } q-1 \nmid k; \\ \quad -1 & \text{if } q-1 \mid k. \end{cases}$$

Lemma 2: Let $f \in F[x_1, x_2, ..., x_n]$, and the degree of $f < n(q-1)$, then $\sum_{c_1 \in F} ... \sum_{c_n \in F} f(c_1, c_2, ..., c_n) = 0$ in *F*.

Proof: By linearity it suffices to consider the monomial $x_1^{k_1} x_2^{k_2} ... x_n^{k_n}$, where

$k_1 + k_2 + ... + k_n < n(q-1)$. From hypothesis, there exists a *j* such that $0 \leq k_j < q-1$.

Thus $\sum_{c_1 \in F} ... \sum_{c_n \in F} c_1^{k_1} c_2^{k_2} ... c_n^{k_n} = (\sum_{c_1 \in F} c_1^{k_1}) ... (\sum_{c_j \in F} c_j^{k_j}) ... (\sum_{c_n \in F} c_n^{k_n}) = 0$ since $(\sum_{c_j \in F} c_j^{k_j}) = 0.$

(Proof of the theorem). Consider the *n* variables polynomial $G = 1 - f^{q-1}$. When $f(c_1, c_2, ..., c_n) = 0$, $G(c_1, c_2, ..., c_n) = 1$.
When $f(c_1, c_2, ..., c_n) \neq 0$, $G(c_1, c_2, ..., c_n) = 0$. Thus *G* counts the number of solutions of $f(x_1, x_2, ..., x_n) = 0$, or $\sum_{c_1 \in F} ... \sum_{c_n \in F} G(c_1, c_2, ..., c_n) = N$. On the other hand, the degree of f $< n$, thus the degree of G $< n(q-1)$, hence $\sum_{c_1 \in F} ... \sum_{c_n \in F} G(c_1, c_2, ..., c_n) = 0$ in *F*. This implies $p \mid N$.

There are variations of the theorem. First a weaker version:

Theorem: Let *p* be a prime number, $f(x_1, x_2, ..., x_n)$ be a polynomial of degree $< n$ with coefficients in *F* and such that $f(0, 0, ..., 0) = 0 \bmod p$. Then $f(x_1, x_2, ..., x_n) = 0$ has a nonzero solution mod *p*.

Now a generalization.

Theorem: Let p be a prime number, and let F_q be the finite field with $q = p^t$ elements. For $i = 1, 2, ...m,$ let $f_i(x_1, x_2, ..., x_n)$ be a polynomial of degree d_i in n variables with coefficients in F_q. Let N denote the number of n-tuples $(x_1, x_2, ..., x_n)$ of elements in F_q such that $f_i(x_1, x_2, ..., x_n) = 0$ for all $i = 1, 2, ..., m.$ If $\sum_{i=1}^{m} d_i < n,$ then $N \equiv 0 \bmod p.$

Proof. Let $G(x_1, x_2, ..., x_n) = \prod_{i=1}^{m} (1 - f_i(x_1, x_2, ..., x_n)^{q-1}).$ As before, if $f_i(c_1, c_2, ..., c_n) = 0$ for all $i,$ then $G(c_1, c_2, ..., c_n) = 1,$ and $G = 0$ otherwise. Hence G counts of the number of roots of the system, or

$$N = \sum_{x_1, x_2, ..., x_n \in F_q} G(x_1, x_2, ..., x_n) = \sum_{x_1, x_2, ..., x_n \in F_q} \prod_{i=1}^{m} (1 - f_i(x_1, x_2, ..., x_n)^{q-1}).$$

Since the degree of f_i is $d_i,$ it follows that

$$\prod_{i=1}^{m} (1 - f_i(x_1, x_2, ..., x_n)^{q-1}) = \sum_{r_1, r_2, ..., r_n} a_{r_1, r_2, ..., r_n} x_1^{r_1} x_2^{r_2} ... x_n^{r_n}$$

is a polynomial of degree at most $(q-1)\sum_{i=1}^{m} d_i$ with coefficients in F_q.

Then

$$N \equiv \sum_{x_1, x_2, ..., x_n \in F_q} \prod_{i=1}^{m} (1 - f_i(x_1, x_2, ..., x_n)^{q-1}) \bmod p$$

$$\equiv \sum_{x_1, x_2, ..., x_n \in F_q} \sum_{r_1, r_2, ..., r_n} a_{r_1, r_2, ..., r_n} x_1^{r_1} x_2^{r_2} ... x_n^{r_n} \bmod p$$

$$\equiv \sum_{r_1, r_2, ..., r_n} a_{r_1, r_2, ..., r_n} \sum_{x_1, x_2, ..., x_n} x_1^{r_1} x_2^{r_2} ... x_n^{r_n} \bmod p$$

$$\equiv \sum_{r_1, r_2, ..., r_n} a_{r_1, r_2, ..., r_n} \prod_{j=1}^{n} \sum_{x_j \in F_q} x_j^{r_j} \bmod p$$

where the summation runs over all n-tuples $r_1, r_2, ..., r_n$ of nonnegative integers such that $\sum_{j=1}^{n} r_j \le (q-1)\sum_{j=1}^{n} d_j < n(q-1).$

This implies for some $j,$ $0 \le r_j < q-1,$ and so $\prod_{j=1}^{n} \sum_{x_j \in F_q} x_j^{r_j} \equiv 0 \bmod p.$ The result follows.

Example: We consider equations in F_5.
Let $f(x,y,z)=x^2+y^2+z^2$. We count the number of 3-tuples satisfying $f=0$. First $(0,0,0)$ is one solution. It is not possible that exactly two variables are zeros. Also it is not possible that all three variables are nonzeros since three choices of 1 and 4 do not add up to zero. So exactly one variable is 0 (three choices), say x is 0, then we have 4 choices of y (nonzero), then 2 choices of z. Altogether we have $3\times 4\times 2=24$ choices. Altogether $1+24=25$ solutions.
Let $g(x,y,z,w)=x^2+y^2+z^2+w^2$. Its degree is 2 and has 4 variables. We count the number of 4-tuples satisfying $g=0$. Now $(0,0,0,0)$ is one possible solution. It is not possible that three of the variables are zeros and one variable nonzero. If exactly two variables are zeros, then we are back to the case $f(x,y,z)=0$ with exactly one zero. Hence there are altogether $6\times 4\times 2=48$ solutions. The "6" corresponds to the choice of the two zeros. If none of the variables are zeros, then we must have two variables whose squares are 1 and two variables whose squares are 4, so altogether $6\times 2^4=96$ choices. The "6" again corresponds to the choices of variables whose squares are 1 and each variable has two choices to have its square equal to 1 or 4. Altogether we have $1+48+96=145$ solutions.
We may as well consider $h(x,y,z)=x+yz$. Clearly $(0,0,0)$ is a solution. Easy to see, it is not possible to have exactly one variable being zero. If exactly two variables are zeros, then they are x,y or x,z (not y,z). If $x=y=0$, then we have 4 choices of z. Likewise $x=z=0$, then we have 4 choices of y. If none of the variable are zeros, we have 4 choices of x, then 4 choices of x and 1 choice of z. Hence altogether $4\times 4=16$ solutions. Altogether $1+4+4+16=25$ solutions.

Example (F_{2^n}). We consider solutions of $x^2+y^2+z^2=0$ in F_{2^2}. The elements of the field are $0,1,\alpha,1+\alpha$ with squares respectively $0,1,1+\alpha,\alpha$. Clearly $(0,0,0)$ is one solution. It is impossible to have exactly two zeros. If one of the variables is 0 (three choices), then there are three choices for the other variable and one choice for the third. If all variables are nonzeros, then they can only be $1,\alpha,1+\alpha$. By permuting them (six choices), we have all nonzero solutions. Hence altogether there are $1+3\times 3+6=16=2^4$ solutions.

Example: For $p\geq 3$, the equation $x^2+y^2=1$ contains non-trivial solutions in $F_p=\mathbb{Z}/p\mathbb{Z}$.

The total number of solutions N in $x^2+y^2-z^2=0$ is divisible by p, by Chevally. If there is no solution satisfying $xyz \neq 0$ in F_p, then there are two possibilities. First if there is no a satisfying $a^2+1=0$, then the solutions are $(0,0,0)$, $(0,n,\pm n)$, $(n,0,\pm n)$, $n=1,2,\dots,p-1$, hence there are
$1+(2p-2)+(2p-2)=4p-3\equiv 0 \bmod p$ solutions or $p=3$.

Now if there is a satisfying $a^2+1=0$ in F_p, then there are $N=6p-5\equiv 0 \bmod p$ solutions, or $p=5$. (Add $2p-2$ solution of the form $x^2+y^2=0$.) Some examples: $2^2+5^2\equiv 1 \bmod 7$; $3^2+5^2\equiv 1 \bmod 11$.

An interesting application of the Chevelly-Warning Theorem.

<u>Theorem</u> (Erdos-Ginzburg-Ziv). Given a set of $2n-1$ integers $a_1,a_2,\dots,a_{2n-1}$ integers, there exists a subset $I\subseteq\{1,2,\dots,2n-1\}$ such that:

(i) $|I|=n.$

(ii) $\sum_{i\in I} a_i \equiv 0 \mod n.$

Proof. First Step. Suppose $n=p$ a prime. Define $f_1(t_1,t_2,\dots,t_{2p-1})=\sum_{i=1}^{2p-1} a_i t_i^{p-1}$ and $f_2(t_1,t_2,\dots,t_{2p-1})=\sum_{i=1}^{2p-1} t_i^{p-1}$. Since $f_1(0,0,\dots,0)=f_2(0,0,\dots,0)=0$ and the sum of their degree $2p-2<2p-1$. Thus by Chevally there exists $0\neq x=\{x_1,x_2,\dots,x_{2p-1}\}$ a nonzero vector such that $\sum_{i=1}^{2p-1} a_i x_i^{p-1}=\sum_{i=1}^{2p-1} x_i^{p-1}=0$. Let

$I=\{i \mid 1\le i\le 2p-1 \text{ and } x_i\neq 0\}$. If $x_i\neq 0$, then $x_i^{p-1}=1$ and $0^{p-1}=0$. Thus $\sum_{i\in I} a_i \equiv 0 \bmod p$ and $|I|=\sum_{i\in I} 1\equiv 0 \bmod p$. But $0<|I|<2p$, hence $|I|=p$.

Second Step (Induction). First easy induction, for any $r\ge 2$, if we have $rk-1$ integers, then there exist $r-1$ pairwise disjoint subsets $I_1,I_2,\dots,I_{r-1}\subseteq\{1,2,\dots,rk-1\}$, each of size k, and such that $\sum_{i\in I_j} a_i \equiv 0 \mod k$.

Take $r=2m$ applied to the given set of $2n-1=2mk-1$ integers, give $2m-1$ pairwise disjoint subsets $I_1,I_2,\dots,I_{2m-1}\subseteq\{1,2,\dots,2n-1\}$, each of size k, and such that for all $1 \le j \le 2m-1$, we have $\sum_{i \in I_j} a_i \equiv 0 \mod k$.

Now for each j as above, put $b_j = \sum_{i \in I_j} a_i$, $b_j' = \frac{b_j}{k}$. We thus have $2m-1$ integers $b_1', b_2', ..., b_{2m-1}'$. Again by induction, there exists $J \subseteq \{1,2,...,2m-1\}$ such that $|J| = m$ and $\sum_{j \in J} b_j' \equiv 0 \mod m$. Finally let $I = \bigcup_j I_j$ (disjoint union), $|I| = km = n$ and $\sum_{i \in I} a_i = \sum_{j \in J} \sum_{i \in I_j} a_i \equiv \sum_{j \in J} kb_j' \equiv 0 \mod km$.

Exercises:

1. (Lagrange) Let p be a prime, and suppose $f(x)$ is a polynomial of degree n not whose coefficients are divisible by p. Then the congruence $f(x) \equiv 0 \mod p$ cannot have more than n incongruent solutions. (Commonly used, proof by induction/division algorithm).
2. (Cauchy-Davenport) Let A and B be subsets of integers, then $A+B$ is defined as the set $\{a+b \,|\, a \in A, b \in B\}$. Now suppose p is a prime number, and let A and B be nonempty subsets of $\mathbb{Z}/p\mathbb{Z}$. Show that $|A+B| \ge \min(p, |A|+|B|-1)$.
3. Use Cauchy-Davenport to prove Chevally.
4. Let p be an odd prime, $r \in F_p$, then $x^2 + y^2 = r$ has solutions in F_p.
5. Let p be an odd prime, the number of solutions of $x^2 + y^2 = 1$ over F_p is given by

$$N = \begin{cases} p-1, & p \equiv 1 \mod 4; \\ p+1, & p \equiv 3 \mod 4. \end{cases}$$

References

A. Adler & J. E. Coury, The Theory of Numbers, Jones and Bartlett, London, 1995.

P. L. Clark, The Chevally-Warning Theorem (Featuring the Erdos-Ginzburg-Ziv Theorem), Preprint.

K. Conrad, Finite Fields, Preprint.

G. Everest & T. Ward, An Introduction to Number Theory, Springer, N. Y. 2005.

N. Koblitz, A Course in Number Theory and Cryptology, Springer, N. Y., 1987.
M. B. Nathanson, Additive Number Theory, Springer, N.Y., 1996.

Aims:
1. To show students some structures are needed to solve problems.
2. To know the names of some important theorems.

November 2015

Extra Exercise

(IMO 1995.6) Let p be an odd prime number. Find the number of subsets A of the set $\{1, 2, ..., 2p\}$ such that

(a) A has exactly p elements, and
(b) the sum of all the elements in A is divisible by p.

Let $\xi = e^{2\pi i/p}$ be a primitive root of unity. The idea is to consider products of p elements within the list $\xi^1, \xi^2, ..., \xi^{2p}$.

Then

$$\prod_{i=1}^{2p}(x-\xi^i) = (x^p-1)^2 = x^{2p} - 2x^p + 1.$$

By Viete's formula applied to the coefficient of x^p, get

$$2 = \sum_{1\le i_1 \le ... \le i_p \le 2p} \xi^{i_1+...+i_p} = \sum_{j=o}^{p-1} n_j \xi^j,$$

Where n_j is the number of subsets $\{i_1, i_2, ..., i_p\}$ of $\{1, 2, ..., 2p\}$ such that

$i_1 + i_2 + ... + i_p = j \text{ (mod } p).$

We look for the number n_0.
But ξ is also a root of the polynomial of degree $p-1$, namely

$$P(x) = n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j.$$

On the other hand, the minimal polynomial of ξ is the cyclotomic polynomial

$$\phi_p(x) = 1 + x + x^2 + ... + x^{p-1}.$$

It follows that they differ by a constant c,

$$n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j = c(1 + x + x^2 + \ldots + x^{p-1}).$$

By setting $x = 1$, we find

$$c = \frac{1}{p}\left(\sum_{j=0}^{p-1} n_j - 2\right) = \frac{1}{p}\left(\binom{2p}{p} - 2\right).$$

Using c, we obtain the value of n_0 :

$$n_0 = 2 + \frac{1}{p}\left(\binom{2p}{p} - 2\right).$$